

3ART12 TOUCH

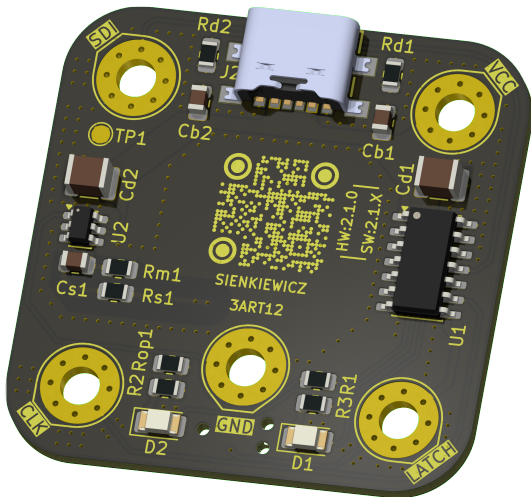
Variant: RELEASED

2026-05-01

Rev 2.1.0

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TOP VIEW



BOTTOM VIEW



NOTES

Comment

Not fitted components are marked as **X**

DRAFT - Very early stage of schematic, ignore details.

PRELIMINARY - Close to final schematic.

CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.

RELEASED - A board with this schematic has been sent to production.

Date: 01-May-2026

DESIGN CONSIDERATIONS

DESIGN NOTE:
Example text for informational design notes.

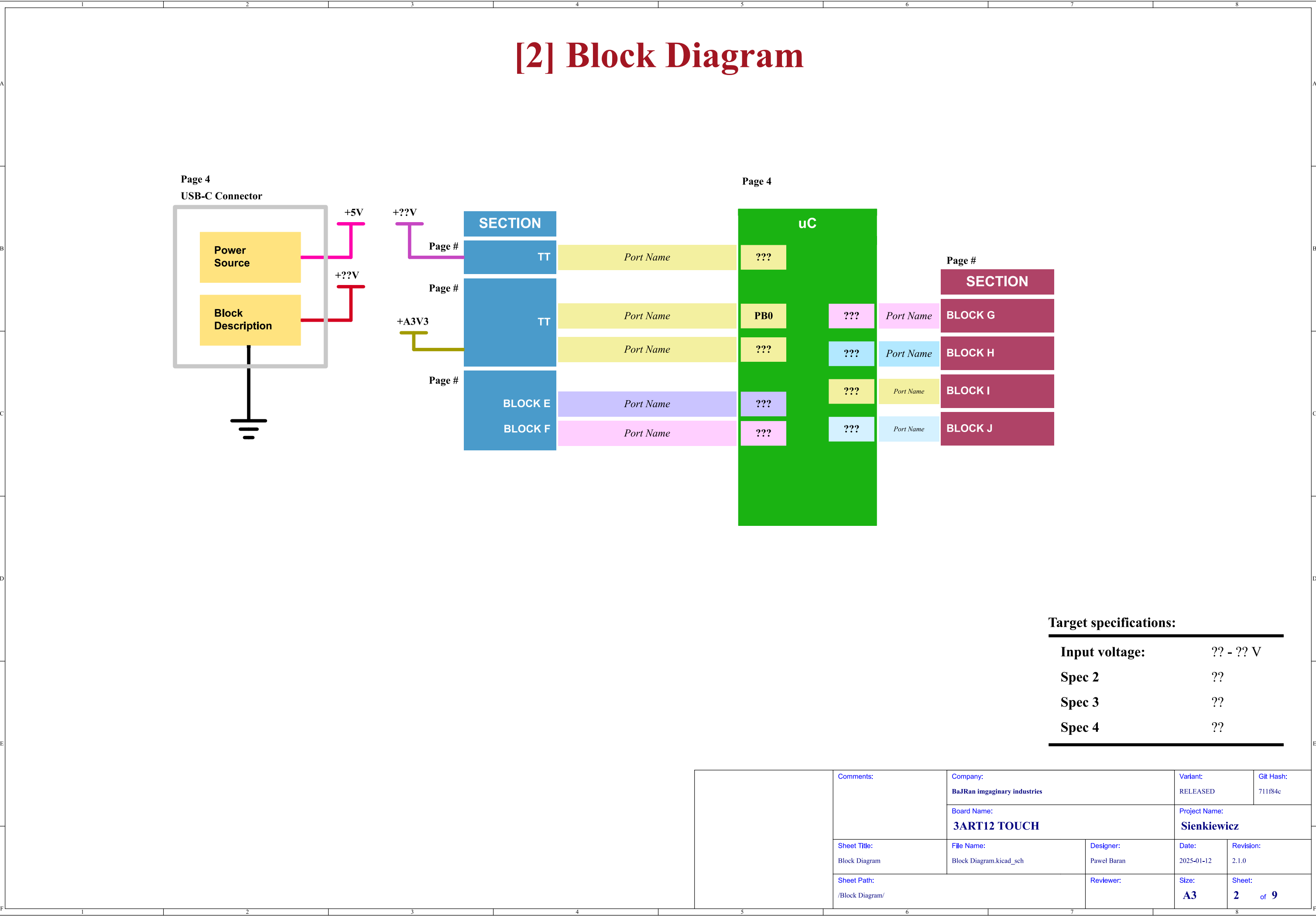
DESIGN NOTE:
Example text for debug notes.

DESIGN NOTE:
Example text for cautionary design notes.

DESIGN NOTE:
Example text for critical design notes.

LAYOUT NOTE:
Example text for critical layout guidelines.

	Comments:	Company: BaJRan imaginary industries	Variant: RELEASED	Git Hash: 711f84c
	Sheet Title:	Board Name: 3ART12 TOUCH	Project Name: Sienkiewicz	
	Sheet Path: /	File Name: Sienkiewicz_TOUCH.kicad_sch	Designer: Pawel Baran	Date: 2025-01-12
			Revision: 2.1.0	
			Size: A3	Sheet: 1 of 9



1

2

3

4

5

6

7

8

A

B

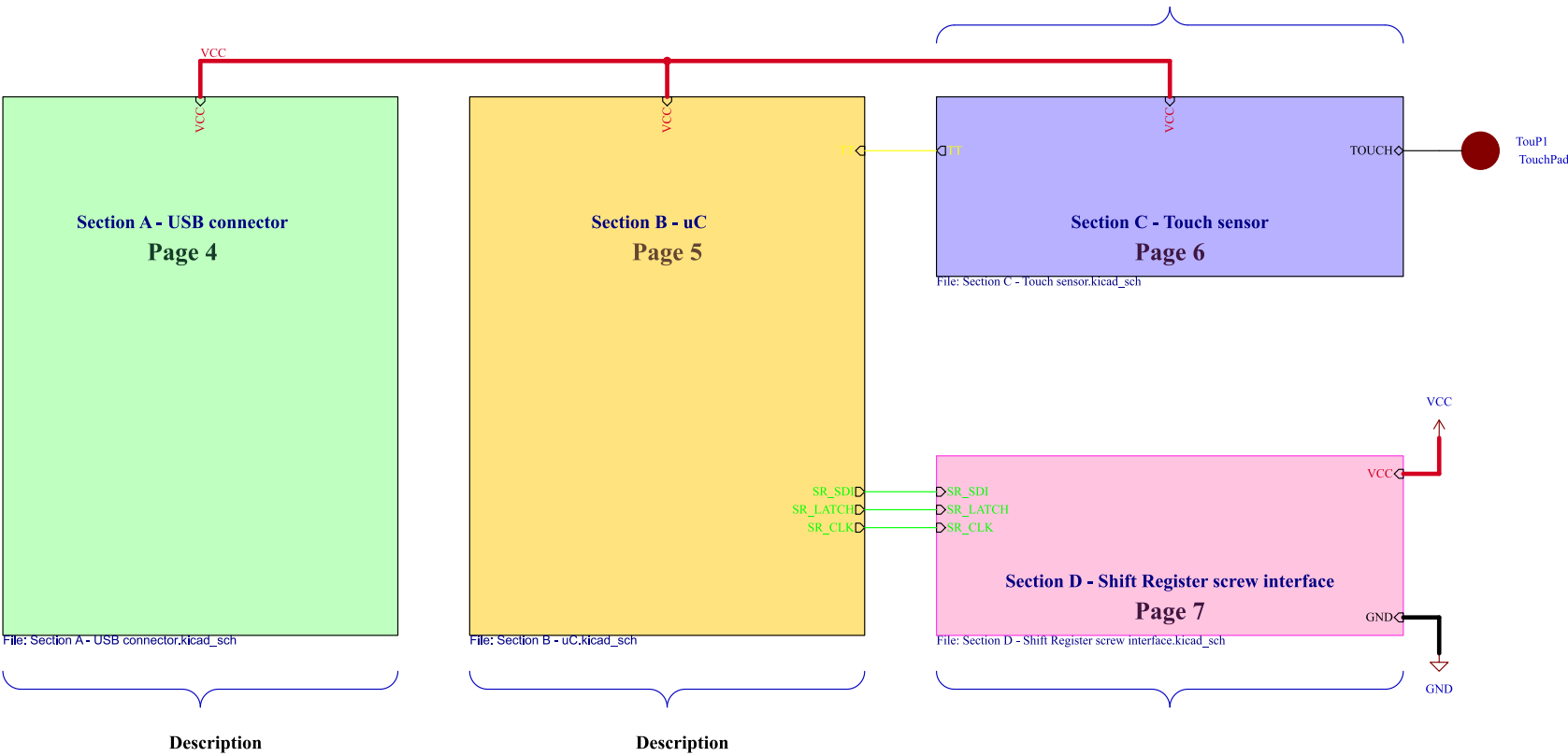
C

D

E

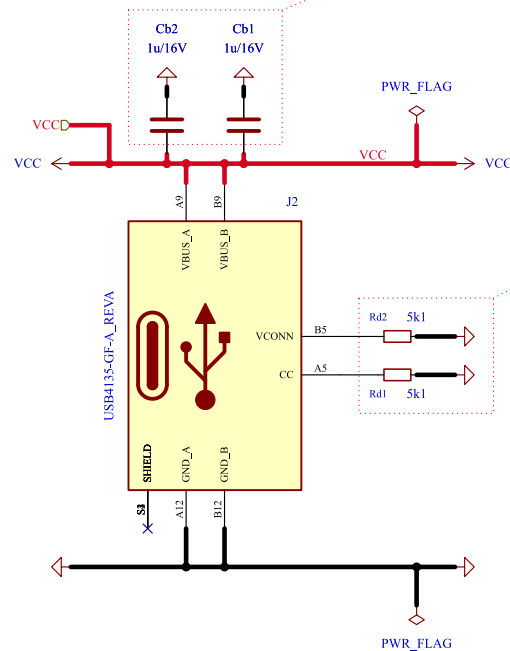
F

[3] Project Architecture



	Comments:	Company: BaJRan  imgaginary industries		Variant: RELEASED	Git Hash: 711f84e	
		Board Name: 3ART12 TOUCH		Project Name: Sienkiewicz		
	Sheet Title: Project Architecture	File Name: Project Architecture.kicad_sch	Designer: Pawel Baran	Date: 2025-01-12	Revision: 2.1.0	
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[4] USB connector



DESIGN NOTE:
Cb1, Cb2
bulk capacitor

Release 2.0
August 2019

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USB Type-C Cable and
Connector Specification

Table 4-2 VBUS Source Characteristics

	Minimum	Maximum	Notes
VBUS Leakage Impedance	72.4 k Ω		Leakage between VBUS pins and GND pins on receptacle when VBUS is not being sourced.
Vbus Capacitance		3000 μ F	Capacitance for source-only ports between VBUS and GND pins on receptacle when VBUS is not being sourced.
		10 μ F	Capacitance for DRP ports between VBUS and GND pins on receptacle when VBUS is not being sourced.

DESIGN NOTE:

Per USB-IF specification, a sink role is established by placing 5.1 k Ω pull-down (Rd) resistors on CC1 and CC2.

Release 2.0
August 2019]

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USB Type-C Cable and
Connector Specification

Table 4-25 Sink CC Termination (Rd) Requirements

Rd Implementation	Nominal value	Can detect power capability?	Max voltage on pin
$\pm 20\%$ voltage clamp ¹	1.1 V	No	1.32 V
$\pm 20\%$ resistor to GND	5.1 k Ω	No	2.18 V
$\pm 10\%$ resistor to GND	5.1 k Ω	Yes	2.04 V

Note:

1. The clamp implementation inhibits *USB PD* communication although the system can start with the clamp and transition to the resistor once it is able to do *USB PD*.

www.usb.org/sites/default/files/USB%20Type-C%20Spec%20R2.0%20-%20August%202019.pdf

LAYOUT NOTE:
blablabla

Comments:

AN5346
STM32G474 Datasheet p.81
J. Pieper ADC investigation

Company:

BaJRan imgaginary industries

Variant:

RELEASED

Git Hash:

711f84c

Board Name:

3ART12 TOUCH

Project Name:

Sienkiewicz

Sheet Title:

USB connector

File Name:

Section A - USB connector.kicad_sch

Designer:

Pawel Baran

Date:

2025-01-12

Revision:

2.1.0

Sheet Path:

/Project Architecture/Section A - USB connector/

Reviewer:

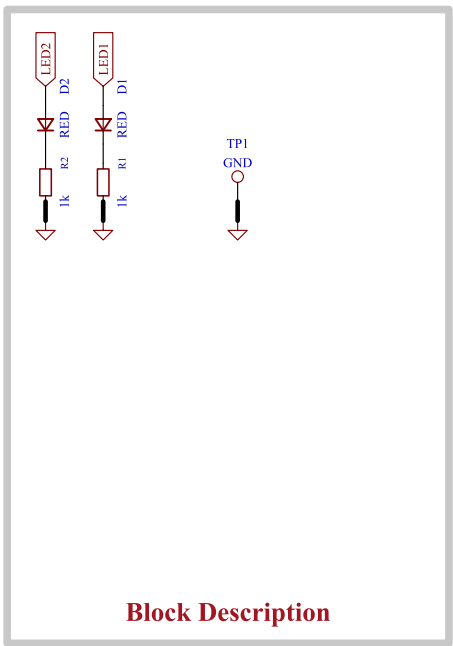
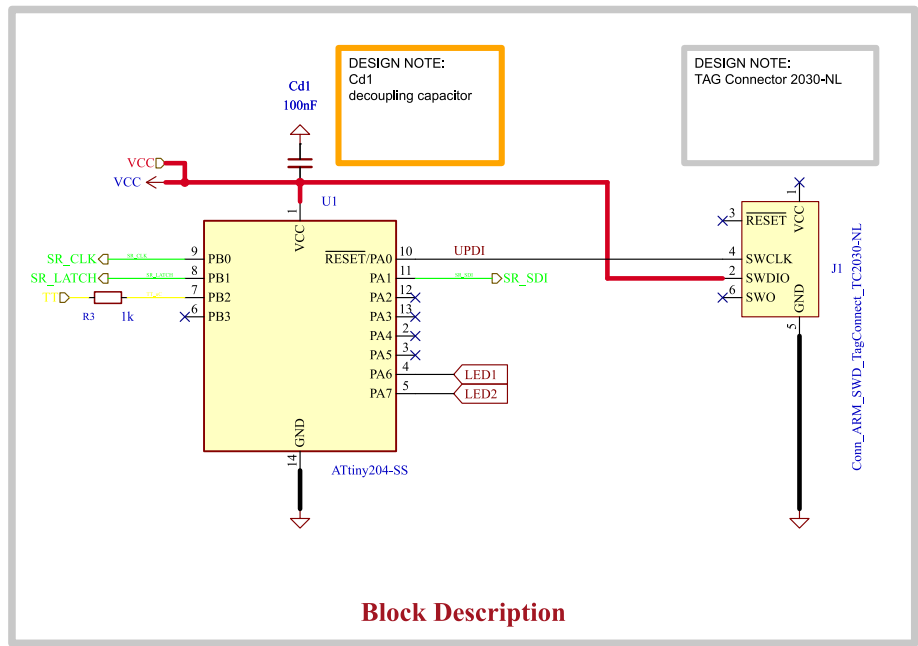
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A4

Sheet:

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[5] uC & prog connector



Description

Ideal for both debugging and production programming of AVR MCUs using the Microchip PICkit 4/5 or SNAP in-circuit debuggers, the 6-pin plug-of-nails™ cable terminates in a 8-pin 0.1" male SIP strip for connection to the debugger. The target board pin connections on the plug-of-nails™ connector are the same as on our TC2030-ICESPI cable. The cable supports programming via PDI, UPDI, aWire, debugWIRE, SPI or TDI.

Note: For AVR processors - DOES NOT WORK WITH PIC MCU

TC2030-PKIT-ICESPI Connections

8 Pin SIP	SPI	TP1	UPDI	PDI	aWire	debugWIRE	6-pin TC2030
1							
2	VTG	VTG	VTG	VTG	VTG	VTG	2
3	GND	GND	GND	GND	GND	GND	6
4	MISO	DAT	PDI_DATA	PDI_DATA	DATA		1
5	SCK	CLK					3
6		RST		PDI_CLK		RESET	5
7	MOSI						4
8							

References:

Flexible I/O worked examples
Flexible I/O source configuration

Comments:

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711f84c

Board Name:

3ART12 TOUCH

Project Name:

Sienkiewicz

Sheet Title:

uC & prog connector

File Name:

Section B - uC.kicad_sch

Designer:

Pawel Baran

Date:

2026-04-16

Revision:

2.1.0

Sheet Path:

/Project Architecture/Section B - uC/

Reviewer:

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A4

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[6] Touch Sensor

DESIGN NOTE:

Table 3-2. Predefined Auto-off Delay (Active High Output)

Vt	Auto-off Delay (t _o)
Vss	Infinity (remain on until toggled to off)
Vdd	15 minutes
OUT	60 minutes

Table 3-3. Predefined Auto-off Delay (Active Low Output)

Vt	Auto-off Delay (t _o)
Vss	15 minutes
Vdd	Infinity (remain on until toggled to off)
OUT	60 minutes

DESIGN NOTE:

Output Polarity Selection

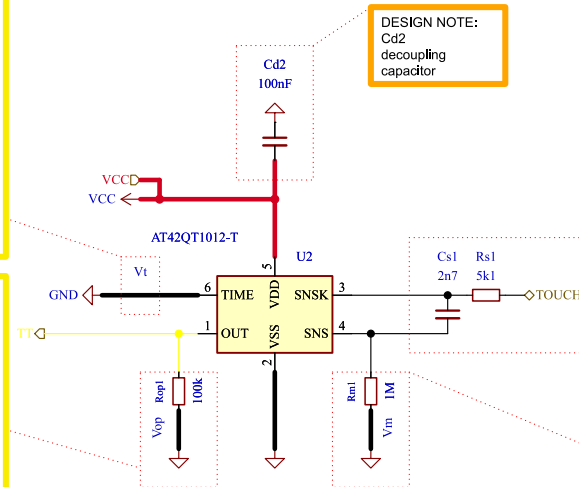
The output (OUT pin) of the QT1012 can be configured to have an active high or active low output by means of the output configuration resistor Rop. The resistor is connected between the output and either Vss or Vdd (see Figure 3-4 and Table 3-1). A typical value for Rop is 100 kΩ.

Table 3-1. Output Configuration

Name (Vop)	Function (Output Polarity)
Vss	Active high
Vdd	Active low

ACTIVE HIGH OUTPUT

AUTO DELAY OFF



DESIGN NOTE:
Cd2
decoupling
capacitor

DESIGN NOTE:
sense circuit
Cs [2nF - 60nF]
Rs [4k7 - 20k]

DESIGN NOTE:

A single 1 MΩ resistor (Rm) is connected between the SNS pin and the logic level Vm to provide three auto-off functions: delay multiplication, delay override and delay retriggering. On power-up the logic level at Vm is assessed and the delay multiplication factor is set to x1 or x24 accordingly (see Figure 3-4 on page 12, Table 3-2 on page 12 and Table 3-3 on page 12). At the end of each acquisition cycle the logic level of Vm is monitored to see if an Auto-off delay override is required (see Section 3.11.5 on page 17). Setting the delay multiplier to x24 will decrease the key sensitivity. To compensate, it may be necessary to increase the value of Cs.

Figure 3-4. Predefined Delay Multiplier

Vm	Auto-off Delay Multiplier
Vss	t _o = 1
Vdd	t _o = 24

Comments:

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Variant:

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Git Hash:

711f84c

Board Name:

3ART12 TOUCH

Project Name:

Sienkiewicz

Sheet Title:

Touch Sensor

File Name:

Section C - Touch sensor.kicad_sch

Designer:

Pawel Baran

Date:

2026-04-14

Revision:

2.1.0

Sheet Path:

/Project Architecture/Section C - Touch sensor/

Reviewer:

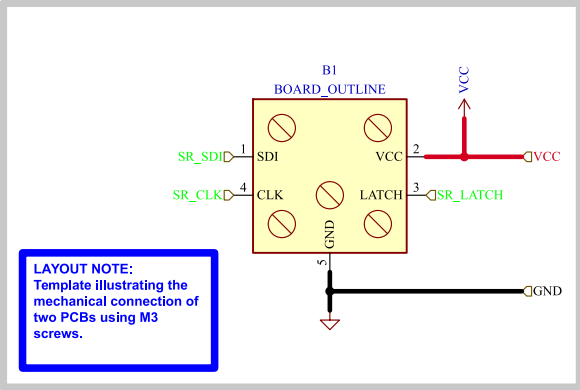
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A4

Sheet:

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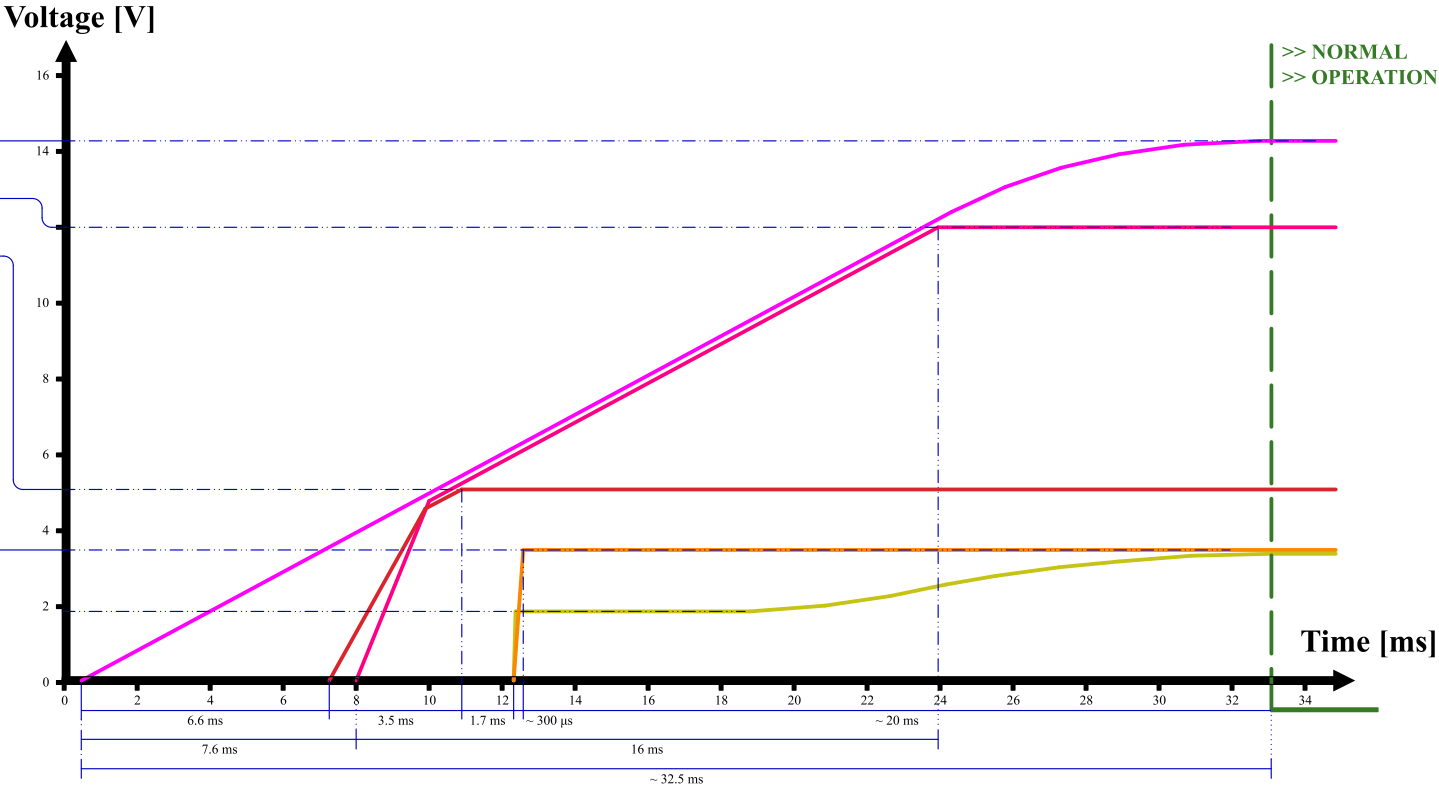
[7] Shift Register screw interface



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				RELEASED	711f84c
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		3ART12 TOUCH		Sienkiewicz	
	Sheet Title:	File Name:	Designer:	Date:	Revision:
	Shift Register screw interface	Section D - Shift Register screw interface.kicad_sch	Pawel Baran	2026-04-16	
	Sheet Path:	Reviewer:	Size:	Sheet:	
	/Project Architecture/Section D - Shift Register screw interface/		A4	7 of 9	

[8] Power - Sequencing

NAME	SOURCE	LEVEL
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%



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	Sheet Title: Power - Sequencing	File Name: Power - Sequencing.kicad_sch	Designer: Pawel Baran	Date: 2025-01-12	Revision: 2.1.0
	Sheet Path: /Power - Sequencing/		Reviewer:	Size: A4	Sheet: 8 of 9

[9] Revision History

Version 2.0.0 - 2026-04-24

- Added
- First 2.0.0 Version of pcb and schematic

Version 2.0.1 - 2026-04-25

- Changed
- Modified silkscreen look, by adding bigger QR-code. small rearangement.

Version 2.0.2 - 2026-04-26

- Changed
- Modified silkscreen look
 - Fabrication layers update

Version 2.1.0 - 2026-05-01

- Changed
- B-B connector maping changes, CLK and SDI swapped (fixes #4)

	Comments:	Company: BaJRan imgaginary industries		Variant: RELEASED	Git Hash: 711f84c
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	Sheet Title: Revision History	File Name: Revision History.kicad_sch	Designer: Pawel Baran	Date: 2025-01-12	Revision: 2.1.0
	Sheet Path: /Revision History/		Reviewer:	Size: A4	Sheet: 9 of 9